

TANMAY SHARMA

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Education

Northeastern University

Master of Science in Cybersecurity

September 2024 – Present

Boston, MA

Coursework: Network Security, Linux Administration, Cryptography, Binary Exploitation, Shell Scripting

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

September 2020 – July 2024

Chennai, India

Coursework: Operating Systems, Computer Architecture, Data Structures, Algorithms, Computer Networks

Experience

CyberspACe securiTy and forensIcs lab (CactiLab)

Research Member

March 2024 – Present

Boston, MA

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Institute for Plasma Research (IPR)

Network Systems Trainee - Cybersecurity

September 2023 – November 2023

Gandhinagar, India

- Designed and deployed a secure, cost-effective network of Raspberry Pi-based access points, creating a dedicated wireless-to-wireless VPN/Wi-Fi solution for an isolated laboratory of 25–30 plasma physicists, enabling seamless and secure intra-lab communication.
- Implemented advanced encryption and security protocols, reducing network vulnerability by 60% and ensuring compliance with stringent research data protection standards, critical for sensitive plasma research experiments.
- Optimized network performance through real-time monitoring and custom configuration, achieving 99% uptime and improving data accessibility for researchers, which enhanced collaboration and accelerated experimental workflows by 30%.

Projects

Secure Instant Messaging System | *Python, SRP, ECDH, TCP Sockets*

January 2025 – April 2025

- Engineered a terminal-based secure chat system leveraging SRP for password-authenticated key exchange, ECDH for secure key derivation, and AES-GCM for end-to-end encrypted messaging, enabling confidential and authenticated real-time communication among users, developed as part of the CY6740 course - Network Security.
- Designed and implemented robust message rate limiting and online user listing features to enhance system integrity and user experience, reducing potential abuse vectors by 80%.
- Developed a socket-based client-server-client architecture supporting encrypted command-based interactions, strengthening the platform's resilience against network-based attacks.

ML Based Phishing Detection System | *Python, Pandas, TensorFlow*

October 2024

- Developed a phishing detection web application leveraging deep learning to analyze URLs and webpage content, achieving 92% detection accuracy and significantly reducing phishing risks for end users.
- Built an interactive web interface enabling real-time URL analysis and instant threat feedback, empowering users to proactively defend against phishing attacks.

Platform for Azure Response & Security Event Correlation | *Azure, AMA, Azure Sentinel*

November 2024

- Designed and deployed a cloud-based SIEM using Azure Sentinel, Log Analytics, and a Windows VM to collect and monitor security events in real time.
- Integrated Windows Security Events via Azure Monitor Agent (AMA), enabling centralized log analysis and scalable threat detection across virtual infrastructure.

Technical Skills

Languages: C/C++, Python, Linux x86 Assembly (32bit/64bit), Bash, PowerShell, Rust (learning), SQL

Security Tools: Metasploit, Wireshark, GDB Debugger, Ghidra, Nikto, Burp Suite, Nmap

Technologies: Cryptographic Protocols, SSL/TLS, IPsec, OAuth, Docker, Git, Linux Administration

Security Domains: Penetration Testing, Vulnerability Analysis, Binary Exploitation, Network Security, Incident Response

Certifications & Achievements

Certifications: NSE 1-8 | Palo Alto Fundamentals of Network and Cloud Security | PNPT by TCM Security (In Progress)
| CompTIA Security+ (In Progress)

Achievements: SimSpace Cyber Cup 2025 - Top 6 Finalist